

World Cup 2026.”

For Sheikh, the lesson extends beyond football. Large sporting events increasingly test not just stadiums but entire urban ecosystems.

“When an event comes to a venue, the entire city becomes a venue,” she says. Perhaps the most visible application of AI has come in one of football’s most contentious areas: officiating.

Offside decisions have sparked debate for as long as the law has existed. At this World Cup, FIFA has introduced a new generation of AI-generated player avatars designed to make those decisions easier to understand.

Before the tournament, all 1,248 players underwent rapid body scans that created highly detailed digital replicas. Combined with semi-automated offside technology and sensor-equipped match balls, the system allows officials to generate lifelike visualisations of incidents for both referees and supporters. The avatars are so detailed that they replicate a player’s body shape and physical appearance rather than relying on generic digital figures.

“When the teams arrive, we scan them,” Sheikh says. “Before, it used to be some cartoon character. You don’t know whether it’s Messi or Ronaldo. Now we are actually building the 3D avatar which replicates their skin tones and physical features.”

The technology, however, cannot eliminate controversy entirely.

“Fans will still curse. Fans are never happy, especially when decisions are not in favour of their team,” Sheikh says with a laugh. “But we wanted to bring more transparency to fans sitting both inside and outside the stadium.”

FIFA and Lenovo hope greater transparency will at least help supporters understand how decisions are reached. The technology has already attracted interest from other sporting bodies looking to replicate similar systems in their own competitions.

Away from the pitch, AI is also helping unlock one of football’s richest yet least exploited assets: its



Competitive equity: For the first time, analysts from Cape Verde can access the same tactical insights available to Spain. Iraq can prepare for France using the same analytical framework employed by football’s traditional powers.

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history.

The sport possesses decades of video footage, photographs and archival material scattered across servers and storage systems. According to Sheikh, AI is helping transform those archives into personalised content, fan experiences and new revenue opportunities.

“The amount of video archives, the amount of content that each team and league has, you won’t believe,” she says. “We are using technology to monetise this, to bring more content to the fan.”

The project required close collaboration between football and technology experts. Lenovo engineers worked alongside FIFA analysts, coaches and operational staff for more than 18 months, refining ideas and adapting systems to the realities of the world’s biggest sporting event.

Sheikh says technology alone was never the challenge.

“You can put the best technology, but if you don’t know how to use it, it is of no use,” she says. “Nothing was done without the experts sitting in

our room. Sometimes we don’t know who’s FIFA and who’s Lenovo. It is such a strong partnership.”

For Lenovo, the World Cup is also a proving ground. The company believes many of the technologies deployed across North America can eventually be adapted for cricket, basketball, baseball and other major sporting events.

“If we can do it for the FIFA World Cup, we can do it for anyone,” Sheikh says. “I am talking to cricket boards, and I am excited to bring all this to the cricket world. Post the World Cup, once we get some breather, that’s the plan, that we go and strategise this.”

Whether it is helping Cape Verde analyse Spain, explaining an offside decision involving Lionel Messi through a digital avatar or coordinating a thunderstorm response in Philadelphia, the technology often fades into the background until it is needed.

The stars still decide the matches. But at World Cup 2026, artificial intelligence and technology are increasingly shaping everything around them.